Session 1

Introduction to Artificial Intelligence

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Artificial Intelligence?

What is Intelligence

- Persepsi (*Perception*)
  - Penginderaan (Sensing) - penglihatan, pendengaran, sentuhan…
  - Pemahaman (Understanding) – penglihatan (vision), pemahaman bicara/ bahasa (speech/language understanding)
- Penalaran (*Reasoning*)
  - Given facts → new facts
- Mempelajari (*Learning*)
  - Meningkatkan kinerja berulang-ulang (berdasarkan pengalaman)
- Adaptasi, kreativitas (*Adaptiveness, creativity, etc*).

What is Artificial Intelligence

- Kemampuan komputer atau robot untuk melakukan tugas-tugas yang biasanya dilakukan oleh manusia karena membutuhkan kecerdasan dan kejelian manusia.
  - Untuk menemukan cara orang berpikir/bertindak secara cerdas
  - Untuk mengembangkan sistem yang melakukan tugas cerdas

What is Artificial Intelligence

- Pada tahun 1955, John McCarthy, salah satu pelopor AI, adalah orang pertama yang mendefinisikan istilah kecerdasan buatan, kira-kira sebagai berikut:
  - *The branch of computer science concerned with making computers behave like humans*
  - *The goal of AI is to develop machines that behave as though they were intelligent*

Systems that think like humans, act like humans,
think rationally, and act rationally

History of AI

Analog-Digital Revolution | *Digital-Semantic Revolution*

The 1st Boom in AI (Late 1950s-1960s)

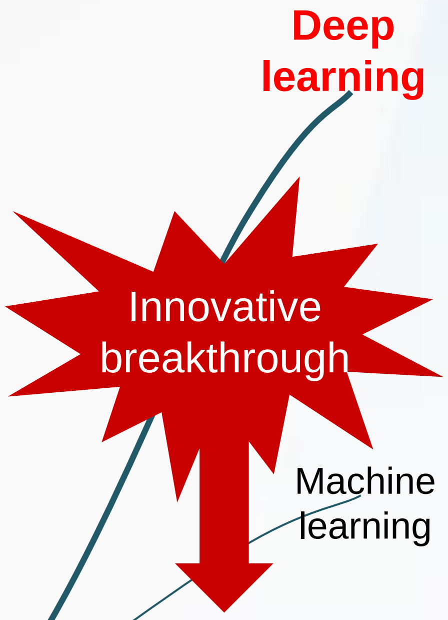
- Upaya pemecahan masalah melalui pencarian dan inferensi menggunakan komputer
- Dengan cepat mereda karena kesulitan dalam menyelesaikan masalah yang kompleks

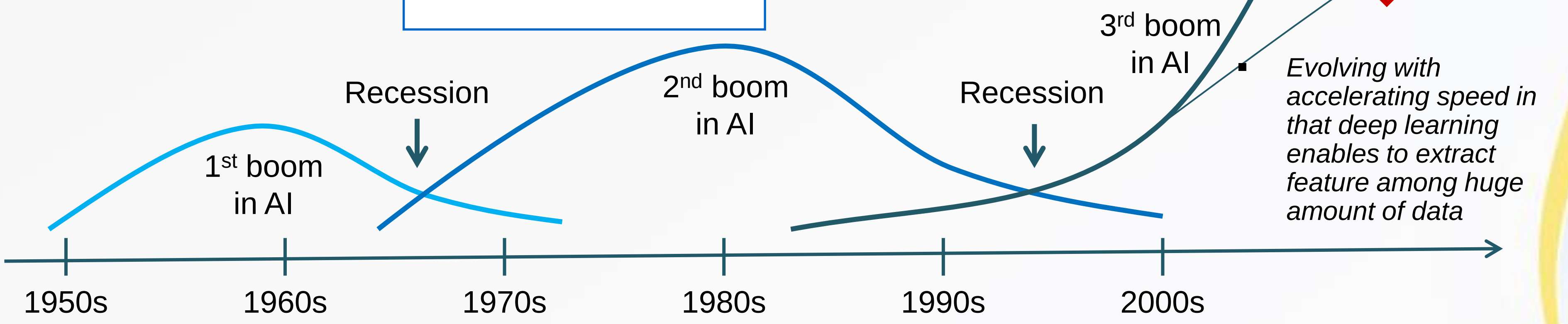
The 2nd Boom in AI (1980s)

- AI yang sangat berpengetahuan
- Zaman keemasan datang dengan pendekatan untuk menjadi cerdas dengan memasukkan 'pengetahuan'/ knowledge' ke dalam komputer sebagai input
- Munculnya 'expert system/sistem pakar'
- Terjadi penurunan karena pengetahuan terlalu besar untuk diwakili dan dikelola

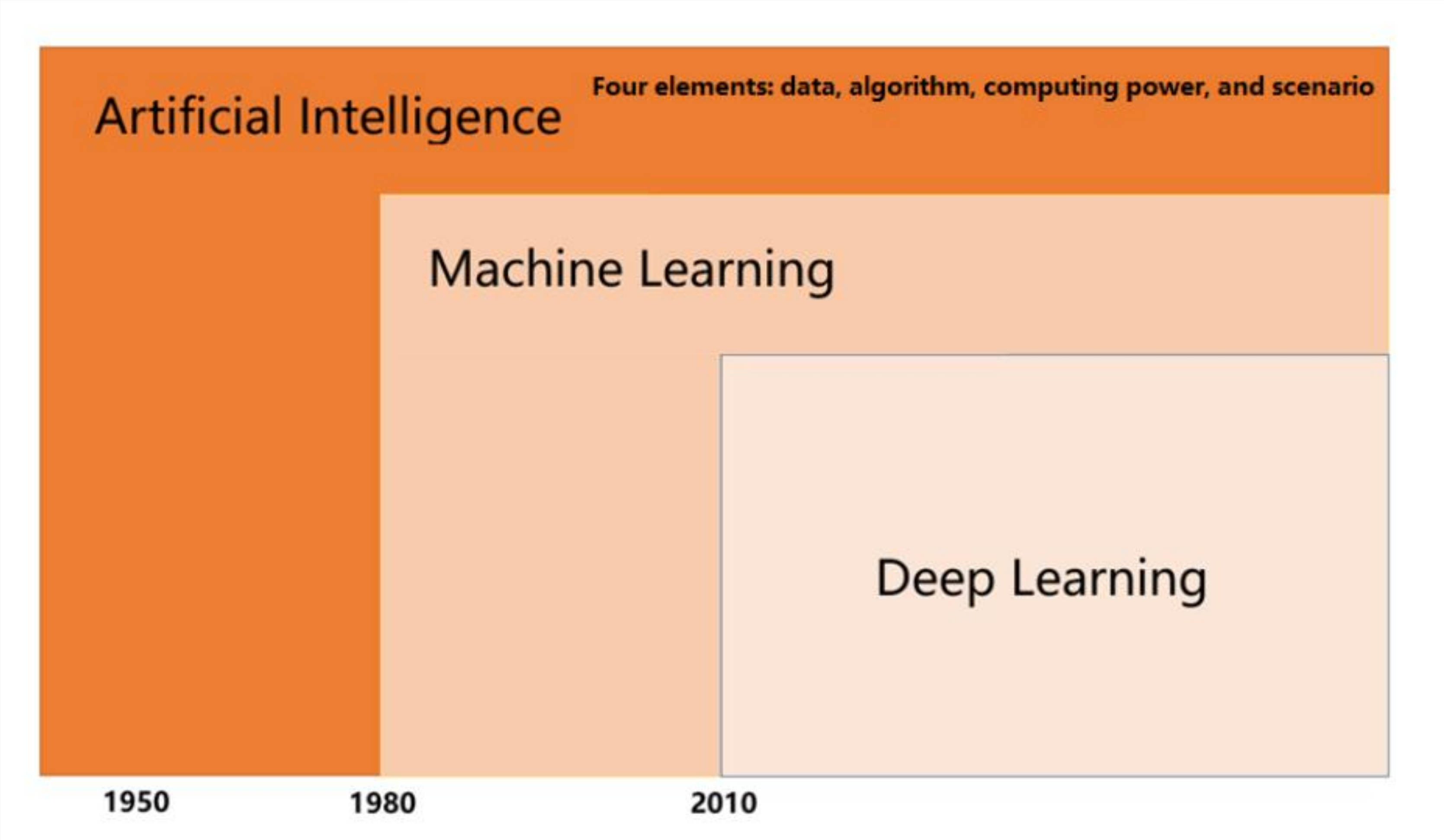
The 3rd Boom in AI (1990s-2000s)

- AI belajar dari data
- 'machine learning/ pembelajaran mesin' menggunakan data yang dikumpulkan sejak kemunculan mesin pencari/ search engine (1990-an)

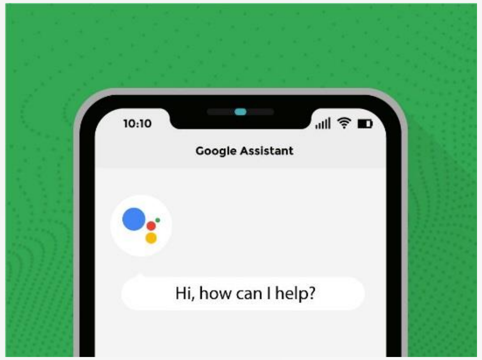




Relationship of AI, ML, and DL

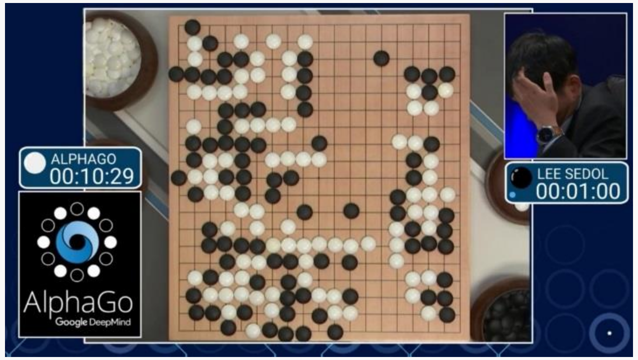


The Emergence of AI



Google Assistant

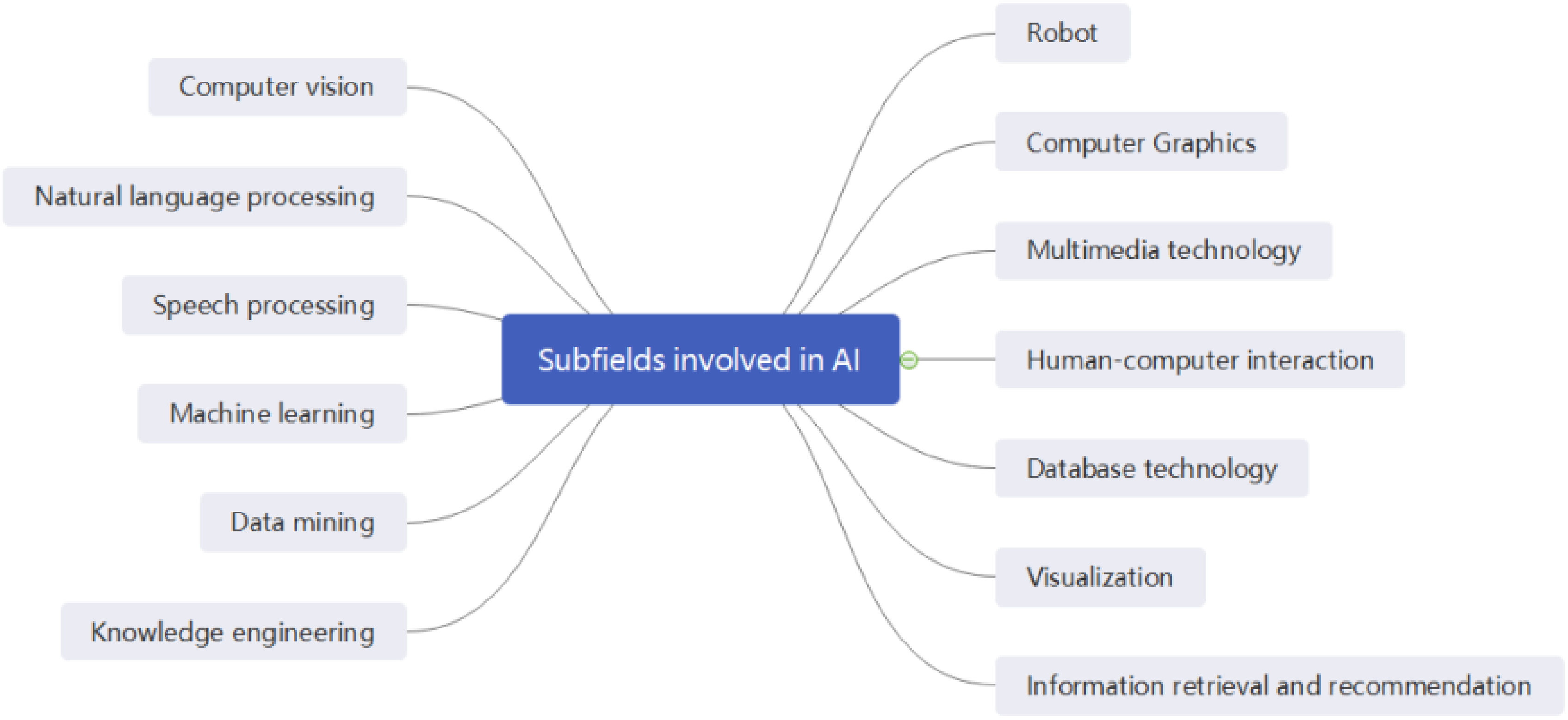
- Virtual assistant developed by Google
- Available on mobile and smart home devices
- Control devices, find information online, control the music.
- May 2018, Google revealed Duplex, that allows it to carry out natural conversations by mimicking human voice,



AlphaGO

- Computer program that plays the board game Go
- It was developed by DeepMind Google
- In 2015, defeated European champion (5-0 wins)
- In 2016, defeated Korean champion (4-1, human's first win)

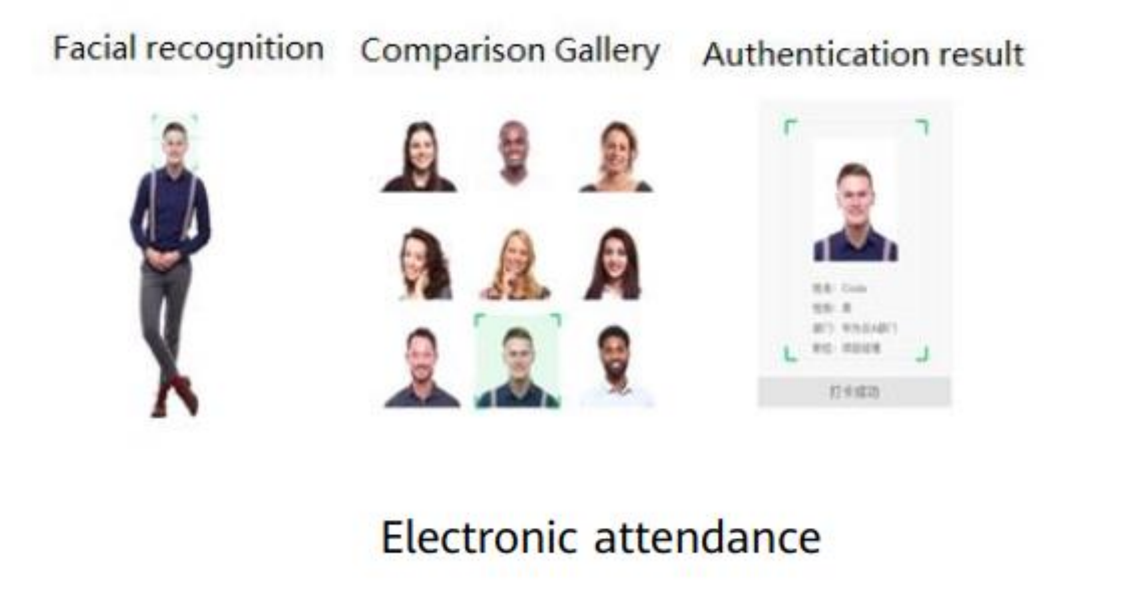
Sub-fields of AI



AI Development Report 2020

Areas/Applications of AI

- Reasoning
  - Fuzzy control
- Expert systems
  - Medical, scheduling,
- Vision
  - Face recognition, autonomous vehicle, object detection
- Speech recognition
  - Voice recognition,
- Natural language understanding
  - machine translation



Areas/Applications of AI

- Machine learning
  - Clustering, classification, reinforcement learning
- Robotics
  - Motion planning, assembly planning
- Game playing
  - Chess, board game Go
- Web agents
  - Recommender systems,

Applications of AI – Smart Home

Control smart home products with voice processing such as air conditioning temperature adjustment, curtain switch control, and voice control on the lighting system.

Implement **home security protection** with computer vision technologies, for example, facial or fingerprint recognition for unlocking, real-time intelligent camera monitoring, and illegal intrusion detection.

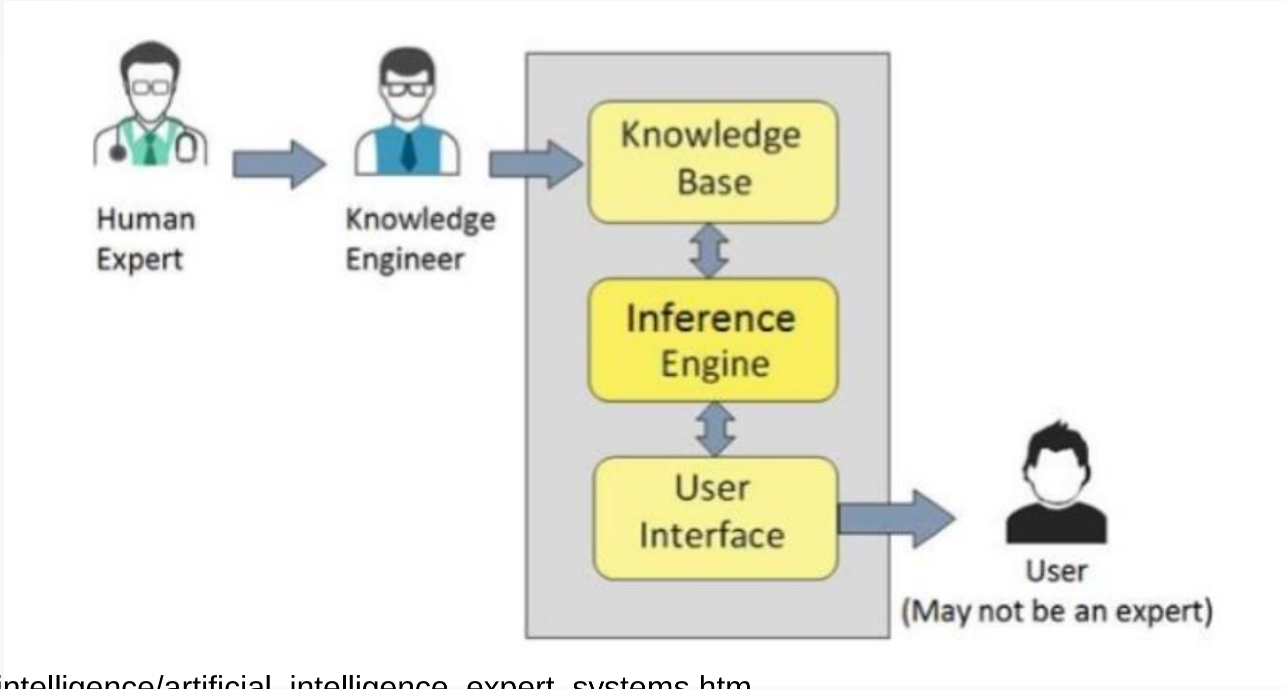
Develop user profiles and recommend content to users with the help of machine learning and deep learning technologies and based on historical records of smart speakers and smart TVs.

Applications of AI – Intelligent Healthcare

- **Medicine mining:** quick development of personalized medicines by AI assistants
- **Health management:** nutrition, and physical/mental health management
- **Hospital management:** structured services concerning medical records (focus)
- **Assistance for medical research:** assistance for biomedical researchers in research
- **Virtual assistant:** electronic voice medical records, intelligent guidance, intelligent diagnosis, and medicine recommendation
- **Medical image:** medical image recognition, image marking, and 3D image reconstruction
- **Assistance for diagnosis and treatment:** diagnostic robot
- **Disease risk forecast:** disease risk forecast based on gene sequencing

Expert Systems

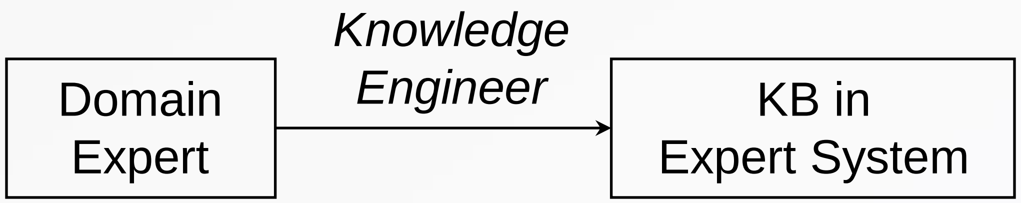
- Expert system
  - Knowledge-based program
  - Provides answers to problems in specific domain



https://www.tutorialspoint.com/artificial_intelligence/artificial_intelligence_expert_systems.htm

Knowledge Engineering

- Knowledge acquisition
  - Building KB by acquiring knowledge from experts
  - → Major bottleneck in building expert system
- Knowledge engineer
  - Select tools
  - Extract knowledge (interview)
  - Build efficient KB



Rule-Based Reasoning

- Represent knowledge as explicit rules

R_1: IF (Outlook=Sunny) AND (Windy=FALSE) THEN Play=Yes

R_2: IF (Outlook=Sunny) AND (Windy=TRUE) THEN Play=No

R_3: IF (Outlook=Overcast) THEN Play=Yes

R_4: IF (Outlook=Rainy) AND (Humidity=High) THEN Play=No

R_5: IF (Outlook=Rain) AND (Humidity=Normal) THEN Play=Yes

Rule-Based Reasoning

- Advantages
  - Use of expert knowledge
    → Good performance in limited domains
  - Explanation is possible
- Disadvantages
  - Need knowledge acquisition
  - Can not handle noisy data

Case-Based Reasoning

- Reasoning from cases
  - Case: examples from past
- Procedure
  - 1. For a new problem, retrieve similar case
  - (based on common feature)
  - 2. Modify the retrieved case
  - (make it suitable for the current situation)
  - 3. Apply the modified case to get the solution
  - 4. Save the new case

Case-Based Reasoning

- Example

customer	sex	age	action
c1	M	40	N
c2	M	20	Y
c3	F	30	N
c4	M	30	Y

c5: F, 20 $\rightarrow$ Y or N?

- Rule-based
  - Rule: if (sex=F) then N. Therefore N
- Case-based
  - c5 is most similar to c3. Therefore N

Case-Based Reasoning

- Issues
  - *Selection of features*
  - *Computing similarity*
- Advantages
  - Akuisisi pengetahuan dapat disederhanakan
  - Menyediakan kemampuan belajar (*learning*) untuk sistem pakar (*expert system*)
- Disadvantages
  - *Lack of explanation capability*
  - *Store/compute trade-offs*

End

Thank You